

Part A: Game Playing AI

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1. Introduction

Artificial intelligence for gaming has been important to the development of AI research both now and in the past. Games have offered structured environments where basic problems like learning, decision-making, and strategic reasoning can be studied in a controlled way, from early systems to modern deep reinforcement agents. Researchers have maintained that games provide a good balance between difficulty and simplicity, making it possible to evaluate AI systems objectively while still presenting difficult problems.

2. Reasons Researchers Create AI Systems that Play Games

2.1 Provide controlled and testable environments

By stating that games offer formal, repeatable, and controllable environments in which intelligence can be scientifically studied, researchers argue for the development of artificial intelligence systems that play games. One of the first research attempts to define game-playing as an artificial intelligence problem is Shannon's paper. According to him, chess is "neither so simple as to be trivial nor too difficult for satisfactory solution" [1]. Shannon believed that games occupied a perfect level of complexity. I agree because it explains why games enable researchers to look at planning, reasoning, and evaluation without the ambiguity of real-world settings, where goals and regulations are frequently ambiguous. Games offer similar benchmarks that make algorithm performance comparable and repeatable because the rules and scoring are set.

2.2 Shows weaknesses in AI methods

The goal of Berliner's research was to create a competitive backgammon-playing program, a game with hidden future outcomes and randomness. Berliner said these games "probably cannot be played by using the chief crutch of most game-playing programs – the look-ahead tree-search" after examining the work [2]. This is significant because it shows how some games highlight the shortcomings of brute-force search. Game-playing AI research pushed academics toward assessing heuristics and learning-based approaches rather than just improving computational capability.

2.3 Offer challenging scenarios

Using Atari video games as test environments, Mnih et al. presented deep reinforcement learning as a technique for learning control rules directly from raw sensory input. According to Mnih et al., these environments are difficult because rewards are "frequently sparse, noisy, and delayed." [4]. This supports the notion that games are purposefully challenging learning exercises. Since these limitations are like those encountered in real-world decision-making systems like robotics and autonomous control, I believe this strengthens the case for game-playing AI.

2.4 Safe and Cost-effective experimentation

Researchers argue that games have advantages since they have extensive experimentation. An investigation was carried out by Del Bosque et al. to investigate the transfer of artificial intelligence approaches from gaming settings to real-world applications. Gaming environments allow AI systems to be assessed "across controlled scenarios that would be costly or unsafe to reproduce in the real world." [8]. It highlights how games act as safe experimental sandboxes, especially for evaluating learning algorithms before implementation in real-world scenarios.

3. Three Application Areas for AI Systems

3.1 Strategic Decision-making

Making strategic decisions in a context of uncertainty is a significant application area. According to Brown and Sandholm, imperfect-information games are relevant to "business strategy, negotiation, strategic pricing, finance, cybersecurity, and military applications," making a clear connection between poker-playing AI and practical fields [5]. Because poker requires long-term strategy, opponent simulation, and probabilistic reasoning, I consider that argument to be fair. These skills apply to real-world conflicts as well as gaming.

Nau explored the wider effects of game-playing AI systems outside of entertainment, highlighting questions about their usefulness outside of games. He argues that these techniques "apply to real problems involving adversarial planning and resource allocation" [6]. This supports the idea that game-playing AI is a foundation for deeper decision-support systems rather than an independent study.

3.2 Robotics and autonomous control

Autonomous control and robotics are the other area. A deep reinforcement learning agent might learn several control policies "directly from high-

dimensional sensory input," as reported by Mnih et al. [4]. This has a lot to do with robotics, as systems must operate in real time and evaluate inconsistent sensor input. This, in my opinion, gives support to the idea that strategies created for gaming can be applied to work requiring physical control, even though more safety precautions are clearly needed.

Studies on applied AI provide more evidence. Many robotic learning strategies "originate from simulated or game-based environments before real-world deployment," [8]. This suggests that games serve as an essential transitional phase between physical systems and theory.

3.3 Entertainment and game design

Playing a game to improve user experience, AI is also directly used in the gaming industry. Lin carried out a research study of the connection between player immersion and artificial intelligence driven game mechanisms. AI "has a very positive impact on the immersion of game players," [9]. I think this application is still important as it shows how AI systems impact human behaviour and involvement, highlighting the growing significance of ethical control. AI for gaming has an impact on game development processes as well, where generative tools speed up iteration and prototyping.

Alharthi researched how professional game design workflows use generative AI tools, focusing on how designers see ownership, efficiency, and originality. According to Alharthi, these tools "accelerate ideation, facilitate prototyping, and automate repetitive processes." [7]. I agree but also think that too much automation might reduce human creativity, which is a problem that needs to be considered in addition to technological performance.

4 Ethical Problems with Game-playing AI

4.1 Opacity and lack of interpretability

Opacity and lack of interpretability are important ethical issue. OpenAI Five, a deep reinforcement learning system trained to play the challenging multiplayer game Dota 2, was studied by Raiman et al. Although the system performed at a high level, the researchers stated that it was difficult to understand how its long-term planning behaviour developed, highlighting the model's "black-box" character [10]. This creates ethical concerns for me because unexplainable algorithms are difficult to check, particularly when comparable methods are applied to real-world contexts outside of gaming.

4.2 Reward Misalignment

Another ethical issue is reward misalignment. Agents can take part in "reward hacking," according to Amodei et al. in which the objective function "can be 'gamed'" through methods that maximize reward in unexpected ways that conflict with the original purpose of the creator [11]. This could result in unfair or inappropriate game strategies. This, in my opinion, brings focus to a more significant ethical problem that AI systems may behave in ways that are technically correct but socially harmful.

4.3 Creativity, Authorship and Player manipulation

Researchers focus on ethical problems related to creativity and authorship. Alharthi claims that designers are worried about "originality, creativity dependency, and the undermining of human-authored content" while using generative AI tools [7]. I believe that depending too much on AI might restrict human creativity and hide accountable for creative decisions. Beyond problems related to creation, AI games could pose ethical risks to players through behaviour impact. "A profound impact on player behaviour" is the result of AI-driven immersion [9]. Increased immersion could be fun, but it runs the risk of encouraging addictive designs if it is not carefully managed.

5 Neural Networks as Game-Playing Agents

Neural networks are strong but not always better. Systems like Chinook used search and endgame databases to achieve world-championship performance before deep learning. The creation of Chinook, a checkers program that finally achieved world-championship performance, was explained by Schaeffer et al. Chinook used "deep search with extensive domain knowledge" instead of learning [3]. This shows that learning is not strictly necessary for good play. Hence, throws doubt on the idea that neural networks are always the best choice for AI that plays games.

Without game-specific tuning, Mnih et al. demonstrated that deep neural networks could perform at a human level in a variety of Atari games [4]. Neural networks, however, play a minor role in other effective systems. Libratus relies more on game-theoretic stability computing than deep learning, as Brown and Sandholm highlighted [5]. This concludes that neural networks perform best when paired with theoretical thinking, planning, and search. It seems that hybrid systems provide the dependable performance.

6 Reliability of References

Because Shannon created the foundation for artificial intelligence, the citation is extremely trustworthy [1]. It has been frequently referenced throughout years of AI study as one of the first formal explorations of game-playing robots, and it still has an impact

on current research on search and decision-making. The paper's conceptual contributions are still important even though it came before modern machine learning techniques, especially when it comes to framing games as scientific tests for intelligence.

Another very trustworthy source is Mnih et al., which was written by scientists at DeepMind, a top AI research company [4]. Deep reinforcement learning techniques were developed in this study, which greatly advanced the field and has been widely cited in both academic and industrial applications. Its reliability and influence on modern reinforcement learning research are enhanced by its reproducible and well-documented experimental technique.

One of the most prestigious and selective peer-reviewed papers is Science, where Brown and Sandholm were published [5]. In this study, a superhuman poker-playing AI is rigorously tested through matches against expert human players. This reference is very reliable and academically strong due to its transparent process, real-world evaluation, and publication venue.

7 Conclusion

Researchers support game-playing AI to improve general intelligence rather than as a goal in and of itself. Although recent developments have been driven by neural networks, the literature shows no single strategy is better in all games. The need for careful examination is further highlighted by ethical issues related to interpretability, reward alignment, immersion, and creativity. One of the most insightful and useful areas of artificial intelligence research is still game-playing AI.

8 References

- [1] Shannon, C. 'Programming a computer for playing chess', *Philosophical Magazine* 41(4) 1950, pp.256–75.
- [2] Berliner, H. 'Backgammon computer program beats world champion', *Artificial Intelligence* 14(2) 1980, pp.205–20.
- [3] Schaeffer, J. et al. 'A world championship caliber checkers program', *Artificial Intelligence* 53:2–3 1992, pp.273–89.
- [4] Mnih, V. et al. 'Human-level control through deep reinforcement learning', *Nature* 518(7540) 2015, pp.529–33.
- [5] Brown, N. and T. Sandholm 'Superhuman AI for heads-up no-limit poker: Libratus beats top professionals', *Science* 359(6374) 2018, pp.418–24
- [6] Nau, D.S. 'AI game-playing techniques: Are they useful for anything other than games? A synopsis of the panel discussion at IAAI-98', *AI Magazine*, 1999, pp.117-118.

- [7] Alharthi, S.A. et al. (2025) 'Generative AI in Game Design: Enhancing Creativity or Constraining Innovation?' (Online) Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12193870/> (Accessed on: 12th January 2025).
- [8] del Bosque, A. et al. 'The role of artificial intelligence in gaming', *Applied Sciences* 15(23) 2024, 12358. (Online) Available at: <https://www.mdpi.com/2076-3417/15/23/12358> (Accessed on: 12th January 2025).
- [9] Lin, H. 'Analysis of the influence of artificial intelligence technology on the immersion of game players', (Online) Available at: <https://arxiv.org/abs/1912.06721> (Accessed on: 12th January 2025).
- [10] Raiman, J., Zhang, S. and Wolski, F. 'Long-term planning and situational awareness in OpenAI Five', (Online) Available at: <https://arxiv.org/abs/1912.06721> (Accessed on: 12th January 2025).
- [11] Amodei, D., Olah, C., Steinhardt, J., Christiano, P., Schulman, J., & Mané, D. (2016). Concrete problems in AI safety. (Online) Available at: <https://arxiv.org/abs/1606.06565> (Accessed on: 12th January 2025).
- [12] Sousa, J.P., Tavares, R., Gomes, J.P. and Mendonça, V. 'Review and analysis of research on video games and artificial intelligence: a look back and a step forward' (Online) Available at: <https://www.sciencedirect.com/science/article/pii/S1877050922007761> (Accessed on: 12th January 2025).
- [13] Hu, C., Zhao, Y., Wang, Z., Du, H. and Liu, J. 'Games for artificial intelligence research: A review and perspectives', (Online) Available at: <https://arxiv.org/html/2304.13269v4> (Accessed on: 12th January 2025).
- [14] Badia, A.P. et al. 'Agent57: Outperforming the Atari human benchmark', (Online) Available at: <https://arxiv.org/abs/2003.13350> (Accessed on: 12th January 2025).
- [15] Mnih, V. et al. 'Playing Atari with deep reinforcement learning', (Online) Available at: <https://arxiv.org/abs/1312.5602> (Accessed on: 12th January 2025).